

Practice Quiz: The Inventor as an Adaptive Agent

Part A: Multiple Choice

1. In Active Inference, what distinguishes an inventor from a non-inventor in how they minimize free energy? A) Inventors have higher intelligence and lower free energy overall B) Inventors emphasize changing the world to match their predictions (active inference) rather than changing their predictions to match the world C) Inventors do not experience free energy because they are creative D) Inventors maximize free energy to generate novel ideas
2. The Wright Brothers' background as bicycle mechanics gave them a generative model that emphasized: A) Maximizing engine power to overcome gravity B) Building the lightest possible airframe C) Dynamic balance and active control during flight D) Static stability through heavy, rigid wing design
3. Spencer Silver at 3M created a weak, reusable adhesive and classified it as a failure. Art Fry saw the same adhesive as a solution for his bookmark problem. What Active Inference concept explains their different responses? A) They had different reaction times B) Their different generative models generated different prediction errors from the same observation C) Silver had higher free energy than Fry D) Fry was a better scientist than Silver
4. An inventor's identity prior refers to: A) Their legal name on patent applications B) Deep beliefs about what kind of inventor they are, which resist updating and shape all other beliefs C) Their most recent invention that establishes their reputation D) The brand identity of their company
5. In a multi-agent invention team, a "shared generative model" means: A) All team members think identically about every aspect of the problem B) The team has agreed on a common set of predictions, goals, and evaluation criteria, even while maintaining individual perspectives C) One team member's model overrides all others D) The team uses the same software tools for modeling

6. The balance between exploitation (refining known solutions) and exploration (trying radically new approaches) in invention is related to which Active Inference concept? A) The Markov blanket B) Expected free energy minimization, which trades off information gain against pragmatic value C) The sensory boundary D) Nested systems
7. A novice inventor has broad but imprecise priors, while an expert inventor has narrow but precise priors. What is a potential disadvantage of the expert's precise priors? A) The expert cannot build prototypes B) The expert may be blind to solutions outside their established model and dismiss unconventional approaches C) The expert always has higher free energy than the novice D) The expert's priors prevent them from having any preferences

Part B: Short Answer and Analysis

1. An experienced architect and a recent art school graduate are teamed up to invent a new type of emergency shelter for disaster relief. The architect's generative model emphasizes structural engineering, building codes, and load calculations. The art graduate's generative model emphasizes user experience, emotional comfort, and creative material use. Describe two specific points where their generative models would likely conflict, and propose one practical mechanism (not just "communicate better") for resolving each conflict productively.
2. Consider the concept of preferred states at three scales for an inventor working on a low-cost water purification device:
3. **Immediate scale:** What might this inventor's preferred state be for today's work session?
4. **Project scale:** What is the preferred state for the completed device?
5. **Identity scale:** What identity-level preferred state might drive this inventor? Then describe a scenario where the identity-level preference overrides a technically optimal project-level decision.
6. Kodak invented the digital camera in 1975 but failed to commercialize it, partly due to the organization's identity as a "film company." Using the Active Inference concept of

resistant priors and free energy minimization, explain (a) why the identity prior was so strong, (b) what evidence should have triggered an update, and (c) what organizational mechanism could have weakened the identity prior enough to allow the evidence to be incorporated. Your answer should explicitly use the terms "generative model," "prediction error," and "prior precision."